Medical Chatbot (End-to-End) Using Natural Language Processing (NLP)

Algorithm Document

1. Dataset Information

Dataset Source

- **Source:** Kaggle
- **Dataset Name:** Disease Symptom Description Dataset

Dataset Link

https://www.kaggle.com/datasets/itachi9604/disease-symptom-description-dataset

Dataset Files

- dataset.csv
- symptom_Description.csv
- symptom_precaution.csv
- Symptom-severity.csv

Dataset Description

The dataset contains disease names, symptoms, disease descriptions, precautionary measures, and symptom severity information. It is used to train the Medical Chatbot to understand user symptoms, predict possible diseases, and provide relevant health guidance and precautionary measures.

Number of Records/Samples

- **4920 Records**

Number of Features (Columns)

- **18 Columns**

Features (Attributes)

- Disease
- Symptom_1
- Symptom_2
- Symptom_3
- Symptom_4
- Symptom_5
- Symptom_6

- Symptom_7
- Symptom_8
- Symptom_9
- Symptom_10
- Symptom_11
- Symptom_12
- Symptom_13
- Symptom_14
- Symptom_15
- Symptom_16
- Symptom_17

Training and Testing Data Split

- **Training Data:** 80%
- **Testing Data:** 20%

Data Format

- CSV (Comma-Separated Values)

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2. Step-by-Step Procedure

1. Start the Medical Chatbot application.
2. Import the required Python libraries such as Pandas, NumPy, NLTK, and Scikit-learn.
3. Load the medical dataset into the Python environment.
4. Read the dataset and inspect it for missing or duplicate values.
5. Clean the dataset by removing unwanted or inconsistent data.
6. Perform text preprocessing on the symptom data:
    - Convert all text to lowercase.
    - Remove punctuation, numbers, and special characters.
    - Tokenize the text into individual words.
    - Remove stop words.
    - Apply lemmatization to obtain the root form of words.

7. Convert the processed text into numerical vectors using the TF-IDF Vectorizer.
8. Split the dataset into training (80%) and testing (20%) datasets.
9. Train the Logistic Regression model using the training dataset.
10. Evaluate the trained model using the testing dataset.
11. Accept a health-related query from the user.
12. Apply the same preprocessing steps to the user input.
13. Convert the processed query into TF-IDF vectors.
14. Predict the possible disease or user intent using the trained model.
15. Retrieve the corresponding disease description and precautionary measures from the medical dataset.
16. Display the disease information, precautions, and health guidance to the user.
17. Continue accepting user queries until the user exits the chatbot.
18. Stop the application.

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3. Logic and Methodology

The Medical Chatbot is developed using Natural Language Processing (NLP) and Machine Learning techniques. Initially, the chatbot accepts symptom-related queries from users through a text interface. The input text is preprocessed using text cleaning, tokenization, stop-word removal, and lemmatization to improve data quality. The processed text is converted into numerical feature vectors using the TF-IDF Vectorizer. A Logistic Regression classifier is trained using the medical dataset to classify user symptoms and identify the most relevant disease. Based on the predicted disease, the chatbot retrieves the corresponding disease description and precautionary measures from the dataset and displays them to the user. This methodology enables the chatbot to provide fast and accurate preliminary healthcare guidance while recommending professional medical consultation whenever necessary.

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4. Model and Techniques Applied

Natural Language Processing Techniques

- Text Cleaning
- Tokenization
- Stop-word Removal
- Lemmatization

Feature Extraction Technique

- TF-IDF (Term Frequency–Inverse Document Frequency)

Machine Learning Model

- Logistic Regression

Programming Language

- Python

Libraries Used

- Pandas
- NumPy
- NLTK
- Scikit-learn
- Streamlit (for chatbot interface)

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5. Process Flow

Start

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Load Medical Dataset

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Read and Clean Dataset

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Text Preprocessing

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Tokenization

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Stop-word Removal

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Lemmatization

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TF-IDF Feature Extraction

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Train Logistic Regression Model

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Model Evaluation

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Accept User Symptoms

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Predict Disease

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Retrieve Description & Precautions

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Display Medical Guidance

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End

Conclusion

The proposed algorithm provides a structured approach for developing a Medical Chatbot using Natural Language Processing and Machine Learning techniques. By combining text preprocessing, TF-IDF feature extraction, and Logistic Regression, the chatbot can analyze user symptoms and predict possible diseases efficiently. The use of a real-world medical dataset from Kaggle improves the chatbot's reliability and prepares a strong foundation for the implementation and source code phases of the project.